

CSE 460: VLSI Design

Lecture 10: Finite State Machines (part 3)

Example-3: Swapping Contents of two registers

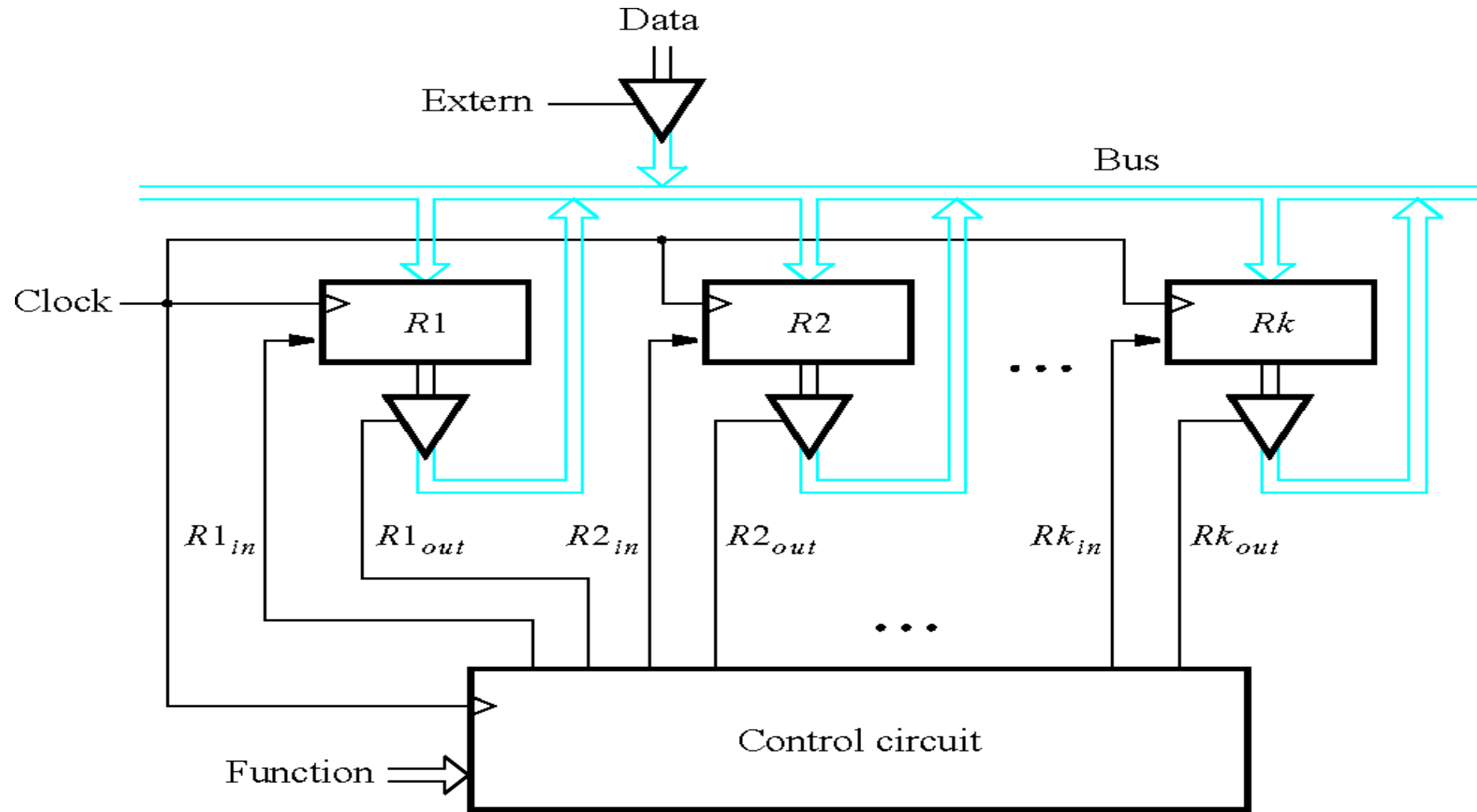


Figure 7.55. A digital system with k registers.

Example-3: Swapping Contents of two registers

Sometimes it is necessary to swap **the contents of two registers**. Typically, this is done by using a temporary location, which is usually a **third** register.

For example, suppose that we want to **swap the contents of registers R1 and R2**. We can achieve this by first transferring the contents of **R2** into the **third register, say R3**. Next, we transfer the contents of R1 into R2. Finally, we transfer the contents of R3 into R1.

Example-3: Swapping Contents of two registers

Task:

1. $R2 \rightarrow R3$
2. $R1 \rightarrow R2$
3. $R3 \rightarrow R1$

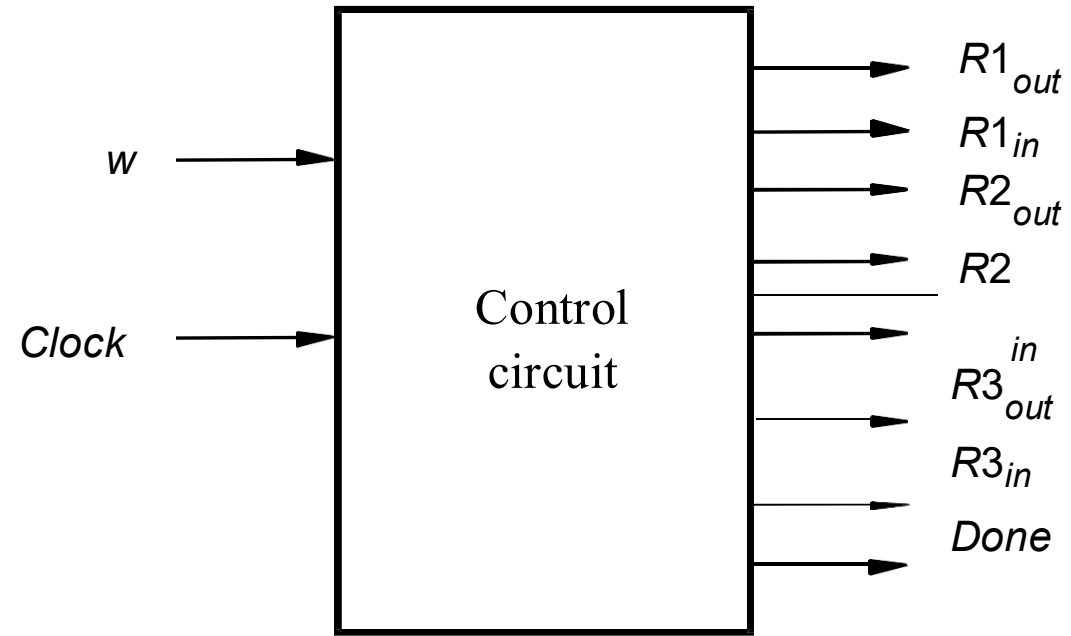


Figure 8.10. Signals needed in this example.

Example-3: Swapping Contents of two registers

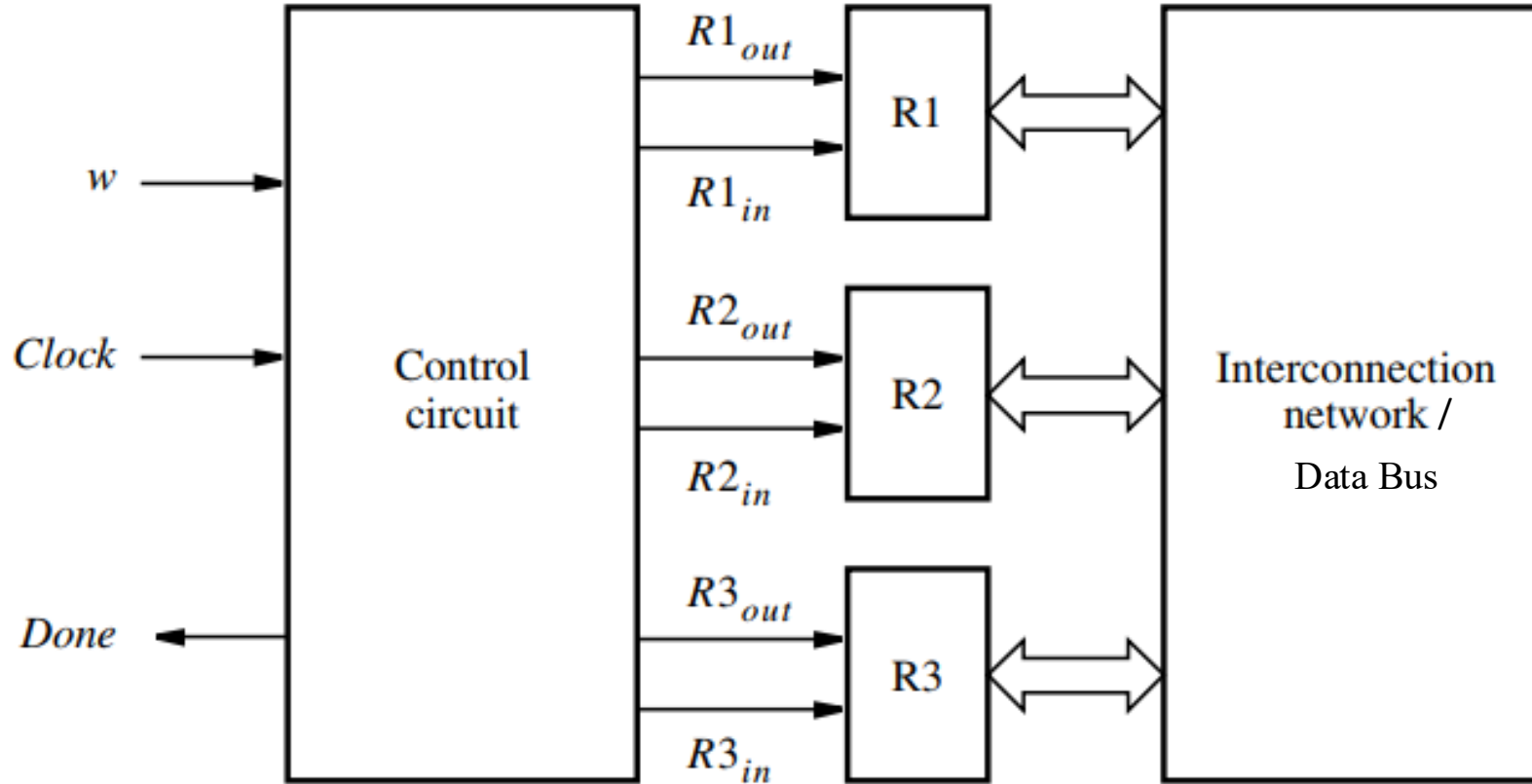
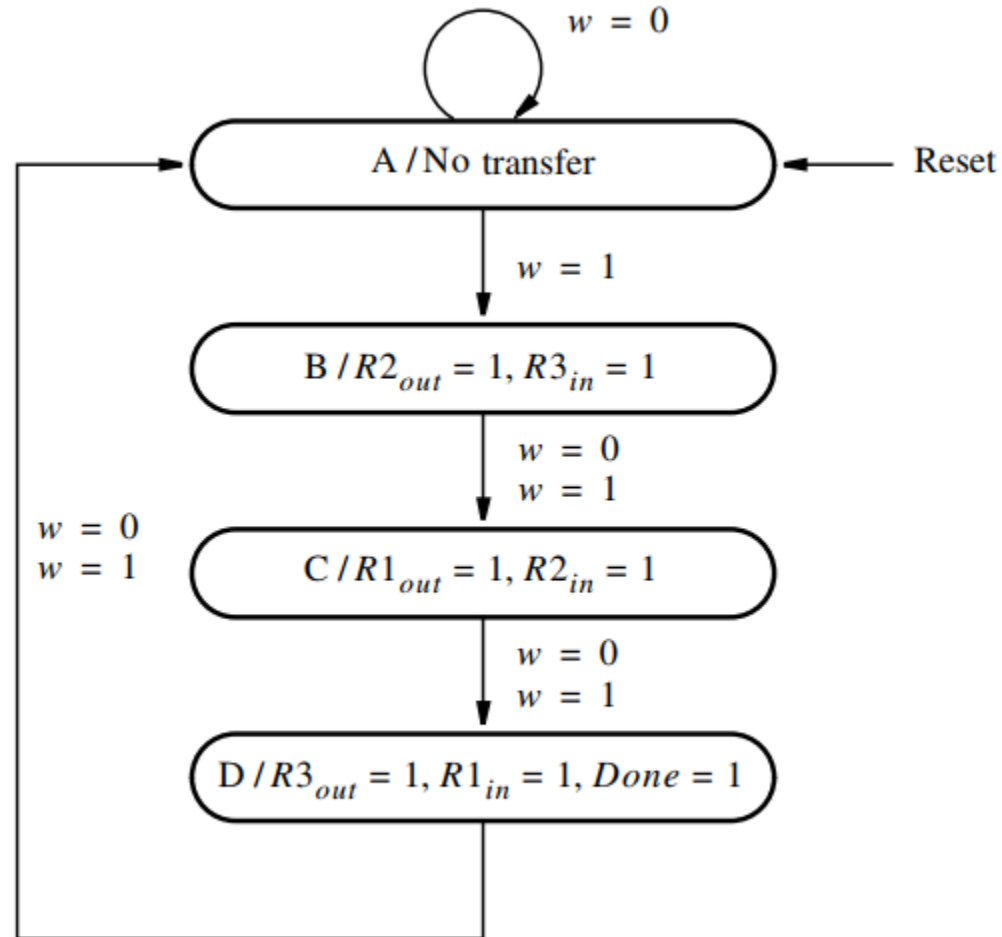


Figure 6.10 The whole system for this example

Example-3: Swapping Contents of two registers (Moore Type)



Once the swapping request arrives ($w=1$), w is neglected until the process ends ($done=1$)

Steps->

- State diagram
- State table
- State assigned table
- K-map
- Circuit

Figure 8.11. State diagram for this example.

Example-3: Swapping Contents of two registers

Present state	Next			state		Outputs						
	w	=	0	w	= 1	$R1_{out}$	$R1_{in}$	$R2_{out}$	$R2_{in}$	$R3_{out}$	$R3_{in}$	$Done$
A	A			B		0	0	0	0	0	0	0
B	C			C		0	0	1	0	0	1	0
C	D			D		1	0	0	1	0	0	0
D	A			A		0	1	0	0	1	0	1

Figure 8.12. State table for Example 8.1.

Steps->

- State diagram
- **State table**
- State assigned table
- K-map
- Circuit

Example-3: Swapping Contents of two registers

	Present state $y_2 y_1$	Next	state	Outputs						
		$w = 0$	$w = 1$							
		$Y_2 Y_1$	$Y_2 Y_1$	$R1_{out}$	$R1_{in}$	$R2_{out}$	$R2_{in}$	$R3_{out}$	$R3_{in}$	$Done$
A	00	00	0 1	0	0	0	0	0	0	0
B	01	10	1 0	0	0	1	0	0	1	0
C	10	11	1 1	1	0	0	1	0	0	0
D	11	00	0 0	0	1	0	0	1	0	1

Figure 8.13. State-assigned table for the sequential circuit in Figure 8.12.

outputs = $f(y_1, y_2)$

$$R1_{out} = R2_{in} = \bar{y}_1 y_2$$

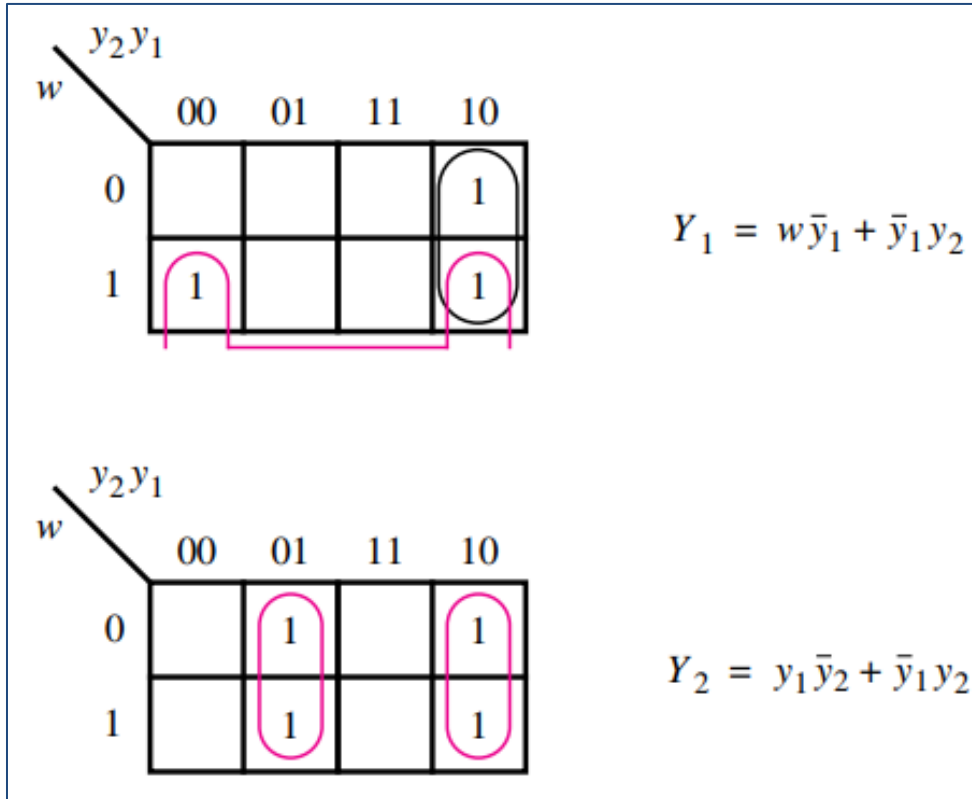
$$R1_{in} = R3_{out} = Done = y_1 y_2$$

$$R2_{out} = R3_{in} = y_1 \bar{y}_2$$

Steps->

- State diagram
- State table
- **State assigned table**
- K-map
- Circuit

Example-3: Swapping Contents of two registers



Steps->

- State diagram
- State table
- State assigned table
- **K-map**
- Circuit

Note that, (Moore type)

- $Y = f(w, y_1, y_2)$
- outputs = $f(y_1, y_2)$

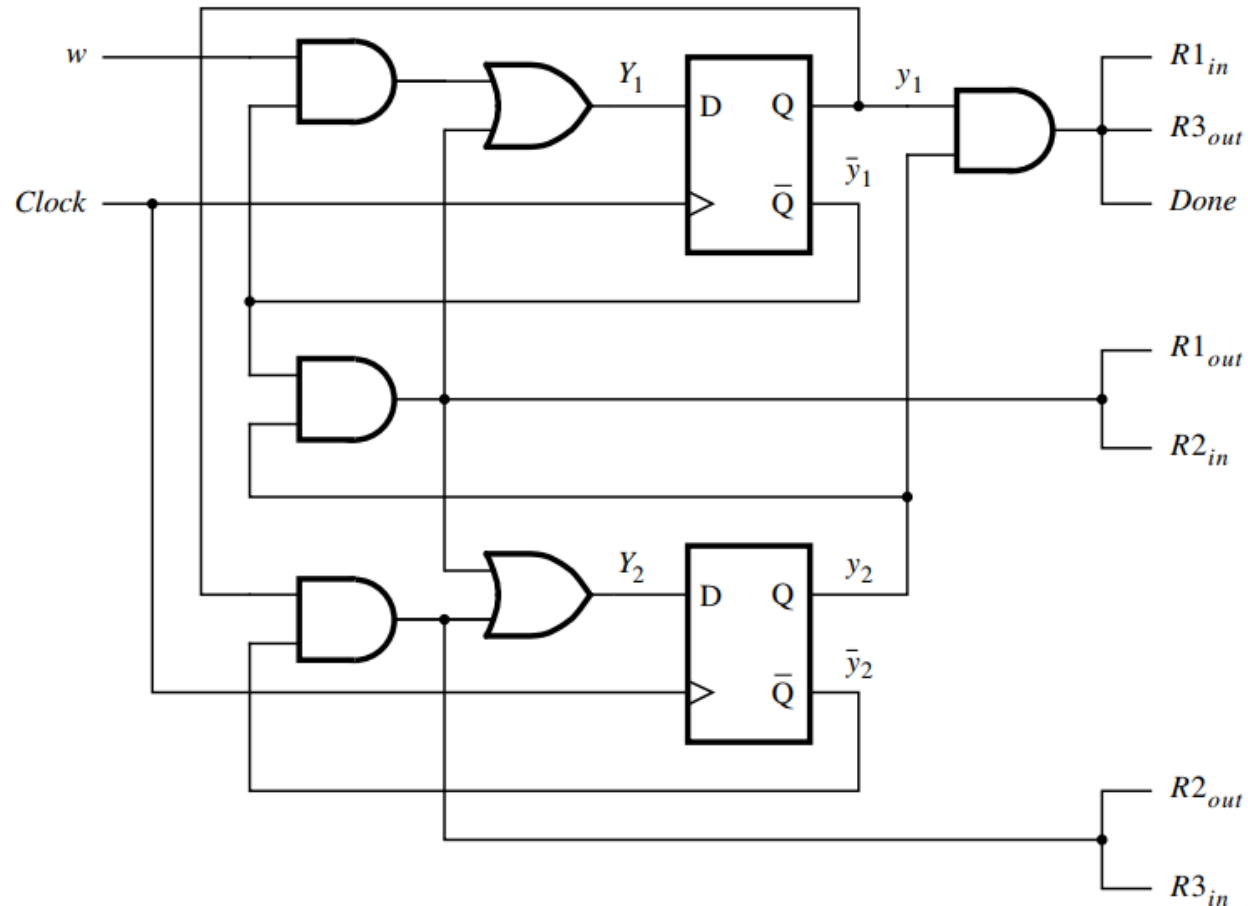
$$R1_{out} = R2_{in} = \bar{y}_1 y_2$$

$$R1_{in} = R3_{out} = Done = y_1 y_2$$

$$R2_{out} = R3_{in} = y_1 \bar{y}_2$$

How many flipflops are required? **Ans: 2**

Example-3: Swapping Contents of two registers



Steps->

- State diagram
- State table
- State assigned table
- K-map
- **Circuit**

Figure 8.15. Final implementation of sequential circuit in Figure 8.13.

Example-3: Swapping Contents of two registers

	Present state $y_4y_3y_2y_1$	Next state		Outputs						
		$w = 0$	$w = 1$							
		$Y_4Y_3Y_2Y_1$	$Y_4Y_3Y_2Y_1$							
A	0001	0001	0010	0	0	0	0	0	0	0
B	0010	0100	0100	0	0	1	0	0	1	0
C	0100	1000	1000	1	0	0	1	0	0	0
D	1000	0001	0001	0	1	0	0	1	0	1

Steps->

- State diagram
- State table
- **State assigned table (One-hot encoding)**
- ~~K-map~~
- Circuit

$$R1_{out} = R2_{in} = y_3$$

$$R1_{in} = R3_{out} = Done = y_4$$

$$R2_{out} = R3_{in} = y_2$$

$$Y_1 = \bar{w}y_1 + y_4$$

$$Y_2 = wy_1$$

$$Y_3 = y_2$$

$$Y_4 = y_3$$

How many
flipflops are
required?

Ans: 4

Example-4: Serial Adder

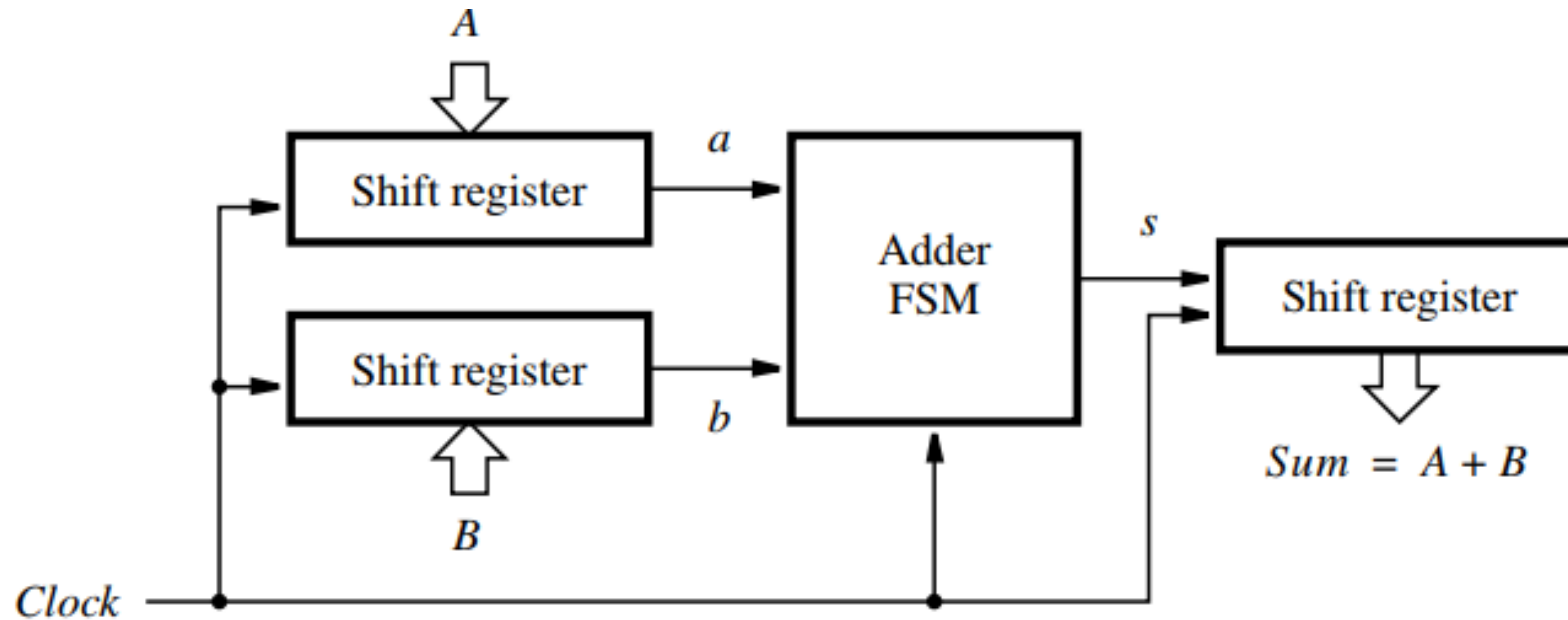


Figure 6.39 Block diagram for the serial adder.

Example-4: Serial Adder (Mealy Type)

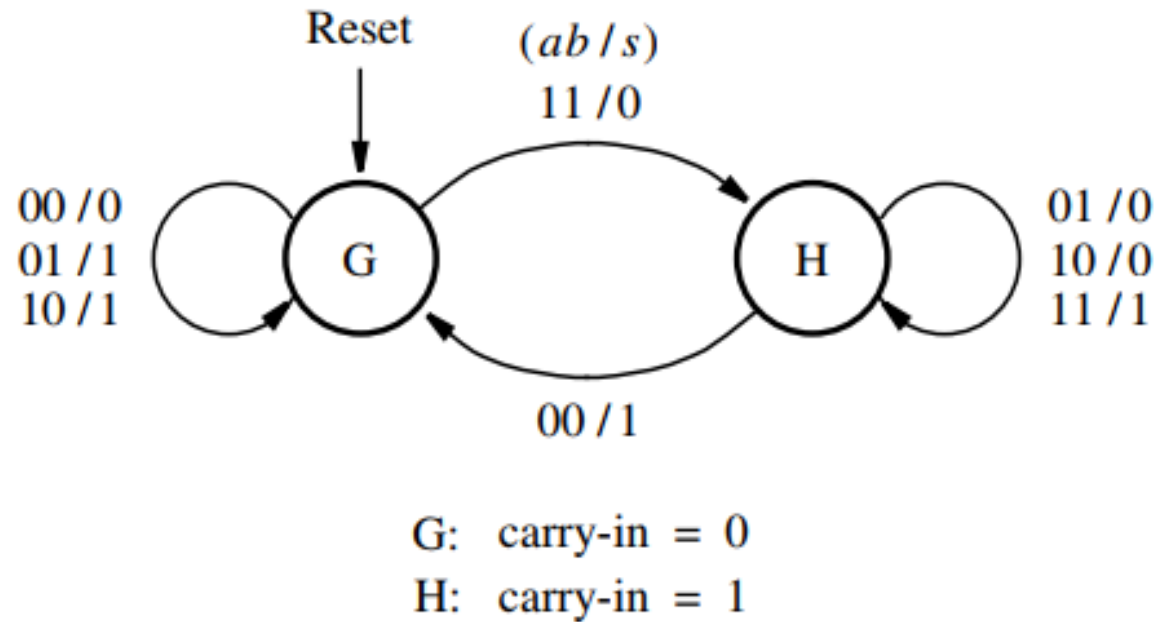


Figure 6.40 State diagram for the serial adder FSM.

Example-4: Serial Adder (Mealy Type)

Present state	Next state				Output s			
	$ab = 00$	01	10	11	00	01	10	11
G	G	G	G	H	0	1	1	0
H	G	H	H	H	1	0	0	1

Figure 6.41 State table for the serial adder FSM.

$$Y = ab + ay + by$$

$$s = a \oplus b \oplus y$$

Present state	Next state				Output			
	$ab = 00$	01	10	11	00	01	10	11
y	Y				s			
0	0	0	0	1	0	1	1	0
1	0	1	1	1	1	0	0	1

Figure 6.42 State-assigned table for Figure 6.41.

Example-4: Serial Adder (Mealy Type)

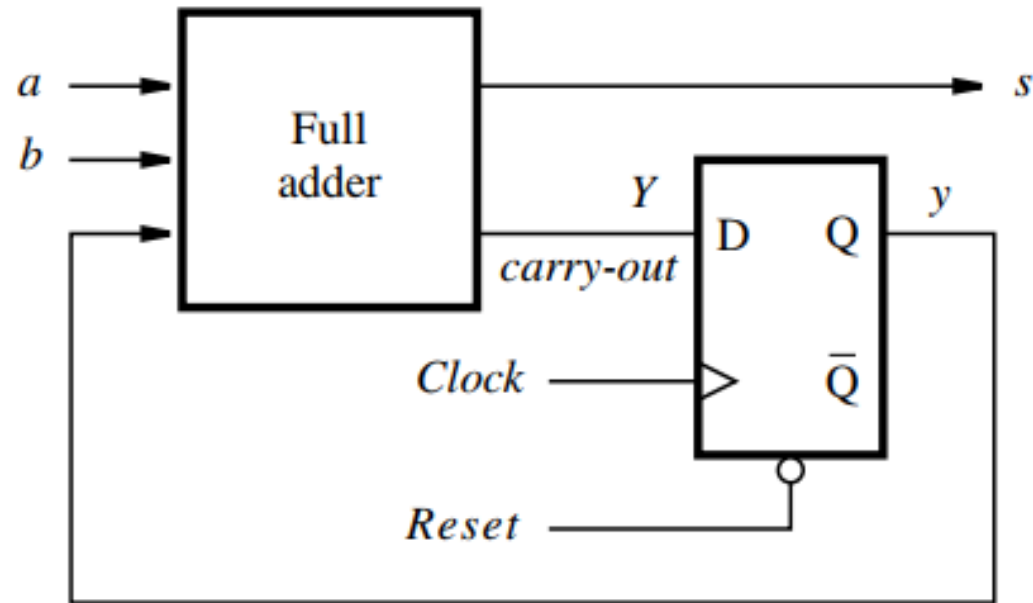


Figure 6.43 Circuit for the adder FSM in Figure 6.39.